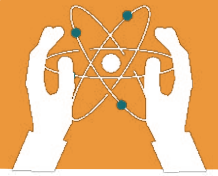


GLOSSARY

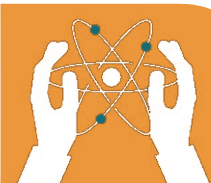
UNIT 01: LIVING AND NON-LIVING THINGS:

- **Adaptation** – A change or trait that helps an organism survive in its environment.
- **Reproduction** – The biological process by which new organisms are produced.
- **Movement** – The ability of an organism to change its position or place.
- **Eukaryotes** – Organisms whose cells have a nucleus enclosed within membranes.
- **Prokaryotes** – Single-celled organisms without a nucleus, such as bacteria.
- **Unicellular** – Made up of only one cell.
- **Multicellular** – Composed of more than one cell, with specialized functions.
- **Diverse** – Showing a great variety or range of differences.
- **Microscopic** – Too small to be seen with the naked eye; needs a microscope.
- **Hibernation** – A state of inactivity and metabolic slowdown during winter.
- **Camouflage** – An adaptation that allows organisms to blend in with their surroundings.
- **Salt Tolerance** – The ability of an organism to survive and thrive in high-salinity environments.



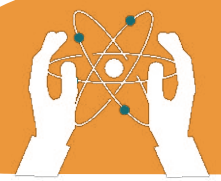
UNIT 02: GASEOUS EXCHANGE:

- **Respiration** – The process of breathing in oxygen and releasing carbon dioxide.
- **Diaphragm** – A muscle located below the lungs that helps in breathing.
- **Bronchi** – Tubes that carry air from the trachea to the lungs.
- **Alveoli** – Tiny air sacs in the lungs where gas exchange occurs.
- **Inhalation** – The act of breathing in air.
- **Exhalation** – The act of breathing out air.
- **Bronchitis** – Inflammation of the bronchial tubes, causing coughing and mucus.
- **Asthma** – A respiratory condition causing breathing difficulty due to narrowed airways.
- **Tuberculosis** – A contagious bacterial infection that affects the lungs.



UNIT 03: NUTRITION IN ANIMALS:

- **Nutrition** – The process of obtaining and utilizing food for growth and energy.
- **Digestion** – The breakdown of food into simpler substances for absorption.
- **Ingestion** – The intake of food into the body.
- **Absorption** – The process of nutrients passing into the bloodstream.
- **Assimilation** – The utilization of absorbed nutrients by body cells.
- **Egestion** – The removal of undigested food from the body.
- **Autotrophic** – Organisms that produce their own food (e.g., plants).
- **Heterotrophic** – Organisms that depend on other organisms for food.
- **Ruminants** – Herbivorous animals with a specialized stomach for digesting cellulose (e.g., cows).
- **Balanced Diet** – A diet containing all essential nutrients in proper proportions.

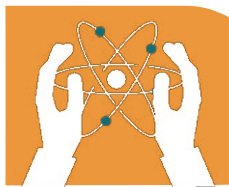


UNIT 04: AIR AND ITS COMPOSITION:

- **Air** – A mixture of gases (mainly oxygen and nitrogen) surrounding the Earth.
- **Atmosphere** – The layer of gases surrounding the Earth, protecting and sustaining life.
- **Global Warming** – The increase in Earth's average temperature due to greenhouse gases.
- **Sailing** – The act of travelling on water using sails powered by wind.
- **Extinguishers** – Devices used to put out fires by releasing a fire-suppressing substance.
- **Inflating** – Filling something with air or gas to make it expand.
- **Particulate** – Tiny solid or liquid particles suspended in the air (e.g., dust, smoke).

UNIT 05: MATERIALS AND THEIR PROPERTIES:

- **Material** – Any substance or matter used to make things.
- **Physical Properties** – Characteristics of a material that can be observed or measured without changing its composition.
- **Hardness** – The ability of a material to resist scratching or indentation.



Texture –The feel or appearance of a surface (e.g., smooth, rough).

Malleable –The ability of a material to be shaped or bent without breaking.

Brittle –A material that breaks or shatters easily under stress.

Conductivity –The ability of a material to transmit heat or electricity.

Density –The mass of a material per unit volume.

Transparency –The property of allowing light to pass through a material.

UNIT 06: ACIDS AND ALKALIS:

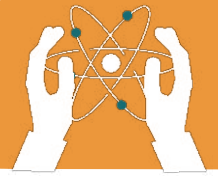
Acid –A substance that donates hydrogen ions (H^+) and has a sour taste.

Alkalis –A substance that donates hydroxide ions (OH^-) and has a bitter taste.

Acetic Acid –A weak organic acid found in vinegar, giving it a sour taste.

Indicator –A substance that changes color to show the presence of an acid or base.

Litmus Paper –A type of indicator paper that turns red in acid and blue in alkalis.



UNIT 07: INTRODUCTION TO PHYSICS:

Nature – The physical world and its natural processes, including plants, animals, and landscapes.

Phenomena – Observable events or occurrences in the natural world.

Physics – The branch of science that studies matter, energy, and their interactions.

Thermodynamics – The study of heat, energy, and work and their transformations.

Mechanics – The branch of physics dealing with motion and forces on objects.

Optics – The study of light and its interactions with matter.

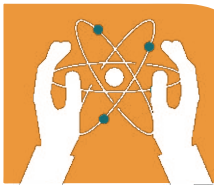
Magnetism – The force of attraction or repulsion due to magnetic fields.

Scientific Methods – Systematic approaches used in scientific investigation, including observation, hypothesis, experimentation, and conclusion.

UNIT 08: FORCES AND THEIR EFFECTS:

Force – A push or pull on an object that can cause it to move, stop, or change direction.

Newton – The unit of force in the International System (SI), symbolized as N.



Magnitude – The size or amount of a force.

Direction – The line or path along which something moves or points.

Contact Force – A force that occurs when two objects are in physical contact (e.g., friction, tension).

Non-Contact Force – A force that acts at a distance without physical contact (e.g., gravity, magnetism).

Gravity – The force of attraction between objects due to their masses.

Friction – A force that opposes motion between two surfaces in contact.

Air Resistance – A type of friction that acts on objects moving through the air.

Static – Refers to an object at rest or forces that are balanced.

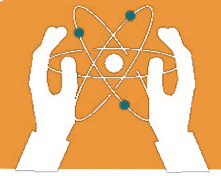
Kinetic – Related to motion; kinetic friction occurs between moving surfaces.

UNIT 09: MAGNETISM:

Magnet – An object that produces a magnetic field and attracts certain metals.

Cobalt – A magnetic metal used in alloys and magnets.

Nickel – A magnetic metal commonly used in coins



Iron – A metal that is commonly attracted to magnets.

Magnetic Material – Substances that are attracted to magnets (e.g., iron, cobalt, nickel).

Non-Magnetic Material – Substances that are not attracted to magnets (e.g., wood, plastic).

Attractive – The force that pulls objects toward each other.

Repulsive – The force that pushes objects away from each other.

Magnetic Field – The area around a magnet where magnetic forces are felt.

Cabinet Latches – Magnetic devices used to keep cabinet doors closed.

Navigation – The process of finding direction, historically using magnetic compasses.